GREENIE

Turning Urban Debris into Economic Value

MARKETPLACE MODULE

PRODUCT REQUIREMENTS DOCUMENT

Team Swach | Sri Krishna College of Engineering & Technology

Version	Status	Date	Team	City
1.0	**Draft**	**Mar 2026**	**Swach**	**Coimbatore**

1. Product Overview

GREENIE's Marketplace Module is the commercial sales arm of the GREENIE Fleet Intelligence platform. GREENIE operates as the sole seller on this platform — collecting C&D waste from contractors at four Transfer Stations across Coimbatore, sorting it using Edge AI, and selling verified, graded secondary materials directly to buyers. This is not a multi-vendor marketplace; it is GREENIE's own digital storefront for its processed material inventory.

Key Distinction from Generic C&D Marketplaces

- GREENIE is the ONLY seller. Every listing originates from GREENIE's own sorted inventory at its Transfer Stations.
- Inventory is auto-synced from the existing Fleet Intelligence system — no manual stock entry needed.
- Buyers see real-time stock, zone origin, and lab-certified grades — not unverified third-party claims.
- This model ensures quality control, regulatory compliance, and brand trust at every transaction.

2. Problem & Business Motivation

2.1 The Revenue Gap GREENIE Currently Faces

GREENIE's Fleet Intelligence system successfully captures, routes, and sorts C&D waste at Transfer Stations. However, the outbound side — selling processed material to buyers — remains informal and relationship-dependent. This creates:

- Delayed off-take: Sorted material sits at Transfer Stations waiting for phone-based buyer negotiations, blocking capacity for new incoming waste.
- Price leakage: Without transparent price discovery, GREENIE is forced to accept below-market prices from repeat buyers who have information leverage.
- Missed buyer segments: Municipal road contractors, individual builders, and small recyclers cannot easily discover or purchase GREENIE material without a personal contact.
- No digital audit trail: Material sale transactions are undocumented, making EPR certificate issuance and CPCB compliance reporting manual and error-prone.

2.2 The Market Opportunity

Metric	Value	Source
India C&D waste generated annually	150–500 million tonnes	MoHUA / Strategic Evaluation Report
Current national recycling rate	~1–2%	MoEFCC estimates
EPR recycling mandate FY 2026–27	50% of generated waste	C&D Waste Management Rules 2025
Coimbatore daily C&D waste	~100–150 TPD	CCMC / GREENIE field data
India C&D recycling market size (2024)	USD 245.40 Billion	Strategic Evaluation Report
Market CAGR (2024–2033)	4.00%	Strategic Evaluation Report
Fine for illegal dumping	INR 5,000–1,00,000	C&D Rules 2025 / Municipal Bylaws
GREENIE gross margin per tonne	~55%	GREENIE Feasibility Slide

3. Goals & Success Metrics

Goal	Metric	Year 1 Target
Accelerate material off-take	Avg. days material sits at station before sale	Reduce from ~14 days to <4 days
Increase outbound revenue	Material Sales GMV through marketplace	INR 2.5 Cr (digital channel)
Reach new buyer segments	Registered buyers across all categories	300+ buyers
Automate EPR documentation	% of transactions with auto-generated WTN + EPR cert	90%

Goal	Metric	Year 1 Target
Improve price realization	Avg. price vs. informal channel baseline	10–15% improvement
Buyer satisfaction	Average order rating	4.3 / 5.0

4. Buyer Personas

GREENIE is the only seller. The following are the buyer personas who will purchase GREENIE's sorted materials through the marketplace.

Municipal Authority (CCMC)	Recycling Plant	Road Contractor	Individual / Small Builder
Bulk concrete rubble for road sub-base. Needs WTN + EPR cert, quantity assurance, and scheduled delivery.	High-purity steel, plastic, glass at scale. Values GREENIE's Edge AI grade guarantees and consistent delivery schedule.	Crushed aggregate for sub-base, WBM. Price-sensitive; prefers nearby Transfer Station zone for lower logistics cost.	Wood, brick, steel for repair/renovation. Needs simple browse, small qty purchase, and quick self-pickup option.

5. GREENIE's Material Catalog (Seller Inventory)

All materials listed on the marketplace are sourced exclusively from GREENIE's 4 Transfer Stations in Coimbatore (North, South, East, West zones). Inventory is auto-populated from the Fleet Intelligence sorting logs.

Material	AI-Sorted Grades	Primary Buyers	Qty Unit	Typical Price Range
Concrete Rubble	Crushed 20mm / Chunked / Reinforced	CCMC, Road Contractors	Tonnes	INR 150–400/tonne
Steel / Rebar	Structural Grade A / Sheet / Scrap	Recycling Plants, Contractors	Tonnes	INR 18,000–28,000/tonne
Wood	Structural Timber / Plywood / Scrap	Individuals, Small Builders	Pieces / Cu.Ft	INR 50–500/piece
Plastic	HDPE / PVC / Mixed	Recycling Plants	Kg	INR 8–25/kg
Glass	Float / Tempered / Wired	Recycling Plants, Glaziers	Sq.Ft / Kg	INR 5–20/kg

Inventory Auto-Sync from Fleet Intelligence

When the Edge AI conveyor at a Transfer Station logs a material batch as 'Sorted & Approved', the Marketplace Inventory Service
automatically creates a new listing draft. The GREENIE admin reviews and publishes it within 2 hours.
Stock quantity is decremented in real-time as orders are placed. Transfer Station capacity is never double-allocated.

6. Feature Requirements

6.1 Public Storefront & Product Discovery (Amazon-inspired)

- Homepage featuring GREENIE's material catalog with category icons (Concrete, Steel, Wood, Plastic, Glass).
- Real-time inventory counter per material category — shows live 'X tonnes available today'.
- Zone filter: buyers can filter by source Transfer Station (North / South / East / West) to minimise their own logistics distance.
- Material detail page: grade description, lot photos from Transfer Station, lab test certificate download (where available), available quantity, price per unit, and origin zone map.
- 'Notify Me' for out-of-stock materials — buyer gets SMS/email when GREENIE restocks that grade.

6.2 Buyer Registration & KYC

- Buyer registration: Name, company (optional for individuals), phone (OTP verify), email, GST number (for B2B), delivery address.
- Buyer type selection at registration: Municipal Body / Recycling Plant / Contractor / Individual — unlocks relevant UI (bulk RFQ for municipal/plant, small-qty checkout for individuals).
- KYC document upload: GST certificate for B2B buyers; Aadhaar OTP for individual buyers.
- Admin approval within 24 hours for institutional buyers (municipal / recyclers) — individuals auto-approved.

6.3 Cart & Ordering

- Add-to-cart with quantity input (min. order enforced per material type: e.g. 5 tonnes for concrete, 10 kg for plastic).
- Zone-aware cart: if buyer adds from multiple zones, system flags logistics complexity and suggests consolidation.
- Delivery mode selection: Self-pickup from Transfer Station (discount applied) or GREENIE-arranged delivery (logistics cost added).
- Delivery scheduling: calendar picker to book pickup/delivery slot — slots fed from Transfer Station availability.
- Order summary: material, grade, quantity, unit price, logistics fee (if any), GST, total — before payment.

6.4 Payments

- UPI, Net Banking, NEFT/RTGS (for B2B bulk orders), and platform-issued pro-forma invoice for government purchase orders.
- Advance payment for standard orders; Purchase Order (PO) upload + credit terms for CCMC and registered municipal buyers.
- GST-compliant tax invoice auto-generated on payment confirmation (e-invoice with IRN/QR code for B2B).
- Partial delivery support: payment split across delivery milestones for large bulk orders (>50 tonnes).

6.5 Order Tracking & Fulfillment

- Order timeline: Confirmed → Transfer Station Preparing → Loaded → In Transit / Ready for Pickup → Delivered → Reviewed.
- Integration with GREENIE Fleet Intelligence: when a truck is dispatched for delivery, the order status auto-updates to 'In Transit' with live tracking link.
- Buyer receives SMS at each milestone (Fleet Intelligence dispatch event triggers notification).
- Digital Proof of Delivery (POD): driver uploads photo at delivery; buyer confirms receipt on app.

6.6 Compliance & Documentation (GREENIE's Regulatory Edge)

- Auto-generated Waste Transfer Note (WTN) per transaction — compliant with C&D Waste Management Rules 2025.
- EPR Certificate generation: each sale contributes to buyer's EPR recycling target; platform auto-computes and issues CPCB-format certificate.
- Blockchain-backed Digital Manifest: transaction hash stored immutably for audit trail (as per GREENIE's proposed EPR architecture).
- CCMC Compliance Report: monthly aggregate export of all transactions for regulatory submission by GREENIE.

6.7 GREENIE Admin Panel

- Inventory dashboard: material stock per zone, pending batches at Transfer Station, sold this week/month.
- Pricing control: GREENIE admin sets and updates price per material/grade — no buyer can modify price (single seller).
- Order management: view all orders, approve PO-based orders, manage delivery slots, assign fleet.
- Buyer management: approve/reject institutional KYC, view buyer order history, blacklist if needed.
- Analytics: revenue by material, by zone, by buyer category; price trend vs informal market baseline.
- EPR & compliance reports: auto-generated monthly CPCB summary; downloadable EPR certificate ledger.

6.8 Ratings & Reviews (Flipkart-inspired — Adapted for B2B)

- After each delivery, buyers can rate: Material Quality (1–5), Grade Accuracy (1–5), Delivery Punctuality (1–5), and leave a comment.
- Reviews are public on the material detail page — builds GREENIE brand credibility with new institutional buyers.
- GREENIE admin can respond to reviews. No anonymous or unverified reviews allowed.

7. Non-Functional Requirements

Category	Requirement
Performance	Page load <2s on 4G; inventory sync from Fleet Intelligence <30s lag; checkout flow <3s.
Availability	99.5% uptime; maintenance 02:00–04:00 IST; Transfer Station sync queue retries on downtime.
Security	AES-256 at rest; TLS 1.3 in transit; JWT auth; DPDP Act 2023 compliant buyer data handling.
Integration	REST API with Fleet Intelligence; event-driven stock updates via webhook on sorting completion.
Mobile	PWA for buyers + admin. Android app in Phase 2. Optimized for low-bandwidth field access.
Compliance	GST e-invoicing (IRN); C&D Waste Rules 2025 WTN format; CPCB EPR digital manifest.
Scalability	Designed for Coimbatore now; architecture supports multi-city replication (Chennai, Bengaluru) in Phase 3.
Accessibility	WCAG 2.1 AA; Tamil + English UI for local buyer access.

8. Integration with GREENIE Fleet Intelligence

Touchpoint	Fleet Intelligence (Existing)	Marketplace (New)	Integration Method
Incoming waste batch	Transfer Station logs sorting completion	Triggers inventory listing draft creation	Webhook: POST /marketplace/inventory/draft
Stock quantity	Real-time sorted batch weight from conveyor IoT	Displayed as 'Available Stock' on product page	REST API poll every 5 min
Delivery dispatch	Fleet assigned + truck GPS tracking active	Order status → 'In Transit' + live	Webhook: POST /marketplace/orders/{id}/dispatch

Touchpoint	Fleet Intelligence (Existing)	Marketplace (New)	Integration Method
		tracking URL to buyer	
Delivery confirmation	Driver logs POD at drop point	Order → 'Delivered'; buyer prompted to review	Webhook: POST /marketplace/orders/{id}/delivered
Admin authentication	Admin login (mail: 123@greenie.ac.in) already exists in Fleet Intel	SSO: same credentials access both systems	JWT token sharing / OAuth2 SSO
Pricing signals	Historical tipping fee and material volume data	Feed into marketplace AI price recommendation engine	Batch export nightly

9. Marketplace Revenue Model

Unlike a multi-vendor marketplace, GREENIE earns the full material sale revenue (no commission deduction). The marketplace layer adds:

Revenue/Cost Component	Description	Est. Impact
Material Sales Revenue	Direct revenue from all marketplace transactions at GREENIE-set prices	Target INR 2.5 Cr Year 1 (digital channel)
Price Premium	Verified grade + digital trust commands 10–15% premium vs informal market	INR 150–600/tonne additional
Faster Inventory Turnover	Reduced avg. storage days from ~14 to <4 means 3× more throughput per station	30–40% capacity utilization gain
EPR Certificate Sales	EPR certs sold to bulk producers on CPCB portal (20% revenue stream)	INR 800–2,500/tonne equivalent
Logistics Margin	If GREENIE arranges delivery, 8–12% margin on freight cost	Add-on per order

10. Phased Rollout Plan

Phase	Timeline	Scope
Phase 1 — Coimbatore MVP	Month 1–3	Inventory auto-sync from Fleet Intel, buyer registration, material catalog, UPI/NEFT checkout, order tracking, WTN generation. All 4 Transfer Station zones.

Phase	Timeline	Scope
Phase 2 — Compliance + Trust	Month 4–6	EPR certificate module, CPCB Digital Manifest (blockchain), buyer ratings/reviews, municipal PO flow, Tamil UI.
Phase 3 — Analytics + Pricing AI	Month 7–10	ML-based price recommendation engine (trained on Fleet Intel data), demand forecasting per material, carbon credit calculation dashboard.
Phase 4 — City Expansion	Month 11+	Replicate platform to Chennai, Bengaluru, Hyderabad Transfer Stations. Multi-city inventory view for pan-India recyclers.

11. Risks & Mitigations

Risk	Impact	Mitigation
Low initial buyer adoption for digital channel	High	Target CCMC first (largest volume buyer); assign one GREENIE field rep for digital onboarding of top 20 buyers.
Fleet Intelligence API instability breaking marketplace inventory	High	Implement async queue with retry; fallback to manual admin stock entry if webhook fails for >30 min.
Pricing disputes (buyers expecting informal market rates)	Medium	Publish price rationale on site (grade cert + WTN included); display 'vs informal market' comparison.
Regulatory changes to EPR certificate format	Medium	Build EPR module with configurable templates; monitor CPCB portal updates monthly.
Seasonal demand dips (monsoon — less construction activity)	Low	Pre-negotiate annual supply agreements with CCMC and recycling plants during peak season.

Appendix — Regulatory Context Summary

The C&D Waste Management Rules 2025 (effective April 1, 2026) create direct commercial demand for GREENIE's marketplace:

EPR Mandate	Year	Why It Benefits GREENIE's Marketplace
Bulk generators must recycle 25% of waste	FY 2025–26	Creates immediate demand for GREENIE's EPR certificates — buyers MUST purchase from certified recyclers like GREENIE.
50% recycling target	FY 2026–27	Doubles the captive market size; GREENIE's marketplace becomes the easiest compliant sourcing channel.

EPR Mandate	Year	Why It Benefits GREENIE's Marketplace
5% recycled material in new construction	FY 2026–27	Contractors building new projects MUST buy from platforms like GREENIE — direct demand for concrete aggregate, steel.
Fine up to INR 1 Lakh for illegal dumping	Immediate	Makes legal sourcing from GREENIE economically and legally superior to all alternatives.